

# Todor Hristov

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## Education

- Sep 2021 – **Computer Science (MSc)**, *ETH Zürich*, Switzerland  
March 2025
  - Overall Grade: 5.38/6.00
  - Received the Jury Award in the Game Programming Laboratory course.
  - Units studied include:
    - Advanced Systems Lab
    - Information Security Lab
    - Program Verification
    - Automated Software Testing
    - System Security
    - Principles of Distributed Computing
    - Design of Parallel and High-Performance Computing
- Sep 2017 – **Computer Science (BSc)**, *University of Bristol*, UK  
July 2020
  - Recipient of The Hargrave Scholarship for Academic Achievement
  - Overall Grade: 81% (First Class Honours)
  - Attended the Algorithms Research Reading Group.
  - Units studied include:
    - Machine Learning
    - Computational Neuroscience
    - Data Structures and Algorithms
    - Concurrent Computing
    - Object-Oriented Programming
    - Software Product Engineering
    - An Introduction to High Performance Computing
- Sep 2012 – **Secondary Education**, *High School of Mathematics "Dr Petar Beron"*, Bulgaria  
June 2017
  - Overall Grade: 5.98/6.00 | Major: Informatics and Mathematics

## Work Experience

- June 2025 – **Software Engineer**, *Huawei*, Zürich, Switzerland  
Present
  - Added Model Context Protocol (MCP) support to the LLM-enabled CodeArts IDE.
  - Designed and optimised algorithms to effectively minimise the usage of the context window.
  - Added MCP UI components for tool call user information, approval buttons, and server monitoring.
  - Validated the MCP functionality using MCP-Bench ([github.com/Accenture/mcp-bench](https://github.com/Accenture/mcp-bench)).
  - Collaborated with cross-functional teams to integrate LLM features into existing IDE pipelines.
  - Gained valuable experience in TypeScript, Prompt Engineering, and System Administration.
- March 2023 – **Research Assistant**, *Oracle*, Zürich, Switzerland  
August 2023
  - Contributed to the development of the Parallel Graph AnalytiX (PGX) platform.
  - Designed and optimised algorithms for efficient computation on large, complex graphs.
  - Implemented new features and enhancements for the PGX platform, leveraging Java and Python.
  - Worked in a supportive, close-knit team to develop robust and scalable software solutions.
- Sep 2020 – **Backend Developer**, *Inexogy Smart Metering (Discovergy GmbH)*, Heidelberg, Germany  
August 2021
  - Developed a Smart Meter Gateway Administrator (SMGWA) for energy data management.
  - Used Go for its development, optimising system performance and integration between components.
  - Leveraged Bash for deployment and automation, streamlining the different processes.
  - Gained hands-on experience in the full development life cycle.
- June 2018 – **Systems Analyst Intern**, *VarSys Ltd.*, Varna, Bulgaria  
August 2018
  - Diagnosed and repaired hardware and software issues, improving technical troubleshooting skills.
  - Effectively communicated technical problems to non-technical clients.
  - Researched Ethereum mining hardware and software for a client project as the primary lead.
  - Assisted in client deployments, recommending solutions and documenting results.

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## Projects

- 2024 **Large Language Models for Email Phishing Training and Education**,  
*Prof. Dr. Srdjan Čapkun, Dr. Daniele Lain, ETH Zürich, Switzerland*
- Master's Thesis with the System Security Group, focused on AI-enhanced email phishing training.
  - Built a modular platform with `React`, `FastAPI`, and `MongoDB`, deployed in three `Docker` containers.
  - Conducted a 201-participant study showing that LLMs enhanced training-page effectiveness.
  - Available on ETH Zürich's Research Collection: [doi.org/10.3929/ethz-b-000695282](https://doi.org/10.3929/ethz-b-000695282)
- 2022 **Seth's Pyramall**, *Game Programming Laboratory, ETH Zürich, Switzerland*
- Developed a 2D platformer in `C#` using the `MonoGame` framework, as part of a six-person team.
  - Recipient of the Jury Award, given by the expert game developers from Studio Gobo.
  - Implemented a grid-based sand simulation, creating realistic flow and piling dynamics.
  - YouTube Trailer: [youtube.com/watch?v=u3z3P-rHndg](https://youtube.com/watch?v=u3z3P-rHndg)
- 2020 **Improving Mouse Detection via Motion Tracking**,  
*Dr. Sion Hannuna, Dr. Neill Campbell, University of Bristol, UK*
- Bachelor's Thesis focused on improving object detection through motion tracking.
  - Applied SORT (tracking-by-detection) to reduce false positives and identify hard-to-detect frames.
  - Performance evaluated against bootstrapping and classical data augmentation methods.
- 2020 **C++ Raytracer**, *Computer Graphics, University of Bristol, UK*
- Developed a raytracer using Simple DirectMedia Layer (SDL) and OpenGL Mathematics (GLM).
  - As part of a two-person team, implemented the following features:
    - Soft shadows
    - Anti-aliasing
    - Mirrors
    - Textures
    - Phong Shading
    - Glass

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## Achievements

- 2020 **Hargrave Scholarship for Academic Achievement**, *University of Bristol, UK*
- Awarded to the top final-year BSc Computer Science student for outstanding academic performance.
- 2018 – 2019 **Top 100 in the World**, *Bloomberg Global CodeCon Finals, UK*
- Represented the University of Bristol at Bloomberg's global coding finals in 2018 and 2019.
  - Ranked Top 100 worldwide, solving eight advanced algorithmic problems in `C++`.
- 2018 **First Place from University of Bristol**, *AI Gaming Mini-Hack, UK*
- Ranked first among University of Bristol participants in an AI game-agent competition.
  - Developed a `Python`-based approximation to the NP-hard Travelling Salesman Problem.
- 2017 **First Place**, *Accessibility Hack 2017, University of Bristol, UK*
- Awarded first place in a group competition focused on creating solutions to enhance accessibility.
  - Developed a Chrome Extension that detects and simplifies complex website text.
  - GitHub Repository: [github.com/LukeStorry/simple\\_read](https://github.com/LukeStorry/simple_read)
- 2015 **Gold Medal**, *National Spring Competition in Informatics, Bulgaria*
- Gold medal (1st place) in one of Bulgaria's most prestigious national programming competitions.
  - Solved three advanced `C++` problems under strict time and memory constraints.

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## Skills

|           |                                                 |                                                       |
|-----------|-------------------------------------------------|-------------------------------------------------------|
| Technical | Programming Languages:                          | C/C++, Python, TypeScript, Java, Go, Bash             |
|           | Parallel Computing:                             | MPI, OpenMP                                           |
|           | Software Development:                           | Test Driven Development, Agile Software Development   |
|           | Tools & Technologies:                           | Git, Docker, L <sup>A</sup> T <sub>E</sub> X          |
|           | Computer Science Fundamentals:                  | Data Structures and Algorithms, Computer Architecture |
| Languages | English (C1)   German (A2)   Bulgarian (Native) |                                                       |

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## Hobbies

Playing table football and simulation video games, such as *Cities: Skylines* and *Minecraft*.